Further practice problems

1 week 3

The topics I covered:

- Laplace matrix, Adjacency matrix, incidence matrix.
- Reduced laplacian and incidence matrix; matrix-tree theorem, Cauchy-Binet.
- There are n^{n-2} spanning trees on n vertices - by Lagrange inversion, and by the matrix-tree theorem.
- Tensor products, graph products, eigenvalues/eigenvectors of adjacency and laplace matrices.
- Cauchy-Binet theorem from linear algebra.
- Spanning trees on the hypercube graph.
- Exponential generating functions, and labeled combinatorial objects.
- counting derangements on n (fixed-point-free permutations).
- operations (addition, multiplication, differentiation, exponentiation/logarithm) on EGFs
- Kruskal's algorithm for minimum spanning tree.
- descents and reduced decompositions/reduced words of permutations (perhaps this was the previous week).

Some questions:

1. let A be the reduced incidence matrix for G. Verify that a maximal minor of A is ± 1 if and only if the corresponding vertices and edges of G form a spanning tree. (This was part of the proof of the matrix-tree theorem, using Cauchy-Binet)
2. A *set partition of* n is an unordered collection of nonempty disjoint sets whose union is $\{1, 2, \ldots, n\}$. Explain why the exponential generating function for set partitions is $G(x) = e^{e^x - 1}$. How could you add a second variable, y, which marks the number of sets in the set partition?
3. Find the exponential generating function for connected graphs on the vertices $1, \ldots, n$.
4. We saw several examples of graphs G for which the number of walks of length n, for all large n, is approximately $C\lambda_1^n$ (where λ_1 is the largest eigenvalue of the adjacency matrix of G and c is some constant). Give an example of a graph G for which this is *not* the case.

5. Suppose $G(x)$ is an EGF that counts some class of labeled objects. What do $\cosh(G(x))$ and $\sinh(G(x))$ count?

6. Prim's algorithm is another algorithm for minimal spanning trees, similar to Kruskal's algorithm. Given a connected graph $G = (V, E)$ and an edge weighting $w : E \to \mathbb{R}$. Start growing a spanning tree T as follows. Repeatedly add the lowest-weight edge which is adjacent to one of the vertices in T and which doesn't create a cycle; stop when T spans G. Why does this create a spanning tree of minimum weight?

7. I could ask the same about Kruskal's algorithm obviously.

8. Let G be a weighted graph, with positive edge weights w_{ij}, and say that the weight of a spanning tree is the product of its edge weights. Describe a matrix G whose determinant computes the sum of the weights of the spanning trees of G. (It may be easiest to imagine that the weights are integers, representing edge multiplicities)

9. Let G be the following directed graph with vertices $\{1, 2, 3\}$, and multiple edges: Each vertex v has two neighbors $a, b \in \{1, 2, 3\}$, with $a < b$; there is one directed edge from v to a, and there are two directed edges from v to b. Count the walks in G of length k which end where they start. Hint: you can actually factor the characteristic polynomial, though you might need to do long division!

10. Describe an algorithm for iterating over all reduced words of a permutation $\pi \in S_n$. Find all reduced words of the permutation 4231 (in one-line notation).

2 week 4

The topics I covered:

- perfect matchings on plane-embedded bipartite graphs; Kasteleyn matrix, partition function of the dimer model.
- spanning trees on the $m \times n$ grid
- Temperley bijection.
- graph transformations which have a predictable effect on the dimer partition function:
  - pendant edge removal
  - double-edge contraction
  - gauge transform
  - spider move
- count of domino tilings on aztec diamond (without evaluating det K)

Some questions. Note that it is somewhat difficult to construct good exam questions about this material - Kasteleyn matrices are kind of huge, and we haven't really done any probability yet. So these are more theoretical and are of the form "did you understand?"

1. Reprove that the above four transforms have a predictable effect on the generating function for perfect matchings of G.

2. Make sure you can run the Temperley bijection.

3. Suppose G is a connected planar graph and that T is a spanning tree of G. Define the corresponding dual spanning tree $T^\vee$ of the dual graph $G^\vee$ and prove that it's a tree.

For all the following questions, let $G = (V, E)$ be a connected planar bipartite graph with the same number of white and black vertices, and suppose G is embedded in the plane with faces F. Let $w : E \to R$ be an edge weight function on G, taking values in the ring R. Some questions:

5. We have observed that when you superimpose two perfect matchings M_1, M_2 on G, you obtain a collection of closed loops and doubled edges (a "double dimer configuration"). Let D be a double-dimer configuration. How many pairs of perfect matchings M_1, M_2 superimpose to give D?

6. Suppose that $\sigma : E \to \{\pm 1\}$ satisfies the Kasteleyn sign condition at all faces of G except possibly F, the external face. Prove that σ also satisfies the Kasteleyn sign condition at F. One approach: prove that if σ is as above and if L is a loop of length $2k$ in G which encloses ℓ vertices, then the product of the signs along L is $(-1)^{1+k+\ell}$ (you need this fact in order to prove that the Kasteleyn matrix works).

7. Show that any G as above has a Kasteleyn sign. The idea is: pick a spanning tree of G, which also means you have implicitly picked a spanning tree $T^\vee$ of the dual graph $G^\vee$. Suppose that $T^\vee$ is rooted at the external face. Argue that you can assign sign $+1$ to all edges in T, and then iteratively assign signs to the edges crossed by $T^\vee$, working from the leaf vertices towards the root.